Fiscal Year:	FY 2023	**Task Last Updated:**	FY 08/03/2022
PI Name:	Mao, Xiao Wen M.D.		
Project Title:	Space Flight Environment Induces Remodeling of Vascular Network and Glia-Vascular Communication in Mouse Retina		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	xmao@llu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	909-558-8373
Organization Name:	Loma Linda University		
PI Address 1:	Radiation Medicine		
PI Address 2:	11175 Campus Street, Rm A1010, Chan Shun Pavilion		
PI Web Page:			
City:	Loma Linda	**State:**	CA
Zip Code:	92350-0001	**Congressional District:**	31
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	10/31/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS Rodent Research-9 NOTE: End date changed to 10/31/2023 per F. Hernandez/ARC (Ed., 2/27/23) NOTE: End date changed to 10/31/2022 per NSSC (Ed., 12/14/21) NOTE: End date changed to 10/31/2021 per F. Hernandez/ARC (Ed., 11/6/18) NOTE: End date changed to 3/31/2020 per F. Hernandez/ARC (Ed., 4/13/18) NOTE: End date changed to 4/30/2019 per F. Hernandez/ARC (Ed., 6/27/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gridley, Daila Ph.D. (Loma Linda University) Pecaut, Michael Ph.D. (Loma Linda University) Sun, Shu-Wei Ph.D. (Loma Linda University)		
Grant/Contract No.:	NNX15AB41G		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	The eye is a unique organ because it is relatively unprotected and constantly exposed to atmospheric oxygen, background radiation, environmental chemicals, and physical abrasion. A recent report shows that more than 30% of the astronauts returning from space were diagnosed with eye problems that can cause headaches and blurry vision. Our Preliminary studies from mice that had been subjected to space flights showed that environmental conditions during space travel lead to oxidative stress and induce adverse microvessel remodeling in the retina. However, our knowledge about the pathophysiological process from adaptive response to irreversible oxidative damage in the retina during and after space flight exposure and the underlying cellular mechanism(s) of these disturbances is very limited and inconsistent. The objectives of this application are to characterize the impact of space flight on female mouse retinal vasculature and tissue remodeling and to study the role of mitochondria in regulating oxidative stress-induced cellular damage and cell-cell interactions in the function of blood-retina-barrier (BRB). Our study will also determine the extent to which an antioxidant metalloporphyrin protects against space flight environment-induced changes in retinal neurovascular coupling and retinal function. The hypothesis will be tested in the specific aims with the space flight experiments on the International Space Station (ISS) and compared to that of ground-based groups using matured female mice. For flight studies, there will be two experiments over two flights, one for in-flight adaptation study on the ISS and one for studying re-adaptation after live animal return to Earth following long-term ISS mission. Selected groups of mice will be treated intraperitoneally with an antioxidant a week before launch and weekly during space flight. Animals will be sacrificed in orbit at 14 and 30 days after launch or 30 days and up to 6 months after return alive on Earth. Together, our unique, integrative, and quantitative activities with advanced MRI (magnetic resonance imaging) imaging techniques and retinal functional testing using visual electroretinography will provide insight into the molecular mechanisms and pathways of space flight condition-induced mitochondrial oxidative damage on retinal BRB integrity, vascular remodeling, and retinal function. The data will provide unique criteria for risks of functional detriments associated with space flight. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward potential countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our proposed study from space flight mission to ISS will provide a unique opportunity to evaluate the effects of space hazards and mechanisms of response and remodeling of the ocular system. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward potential countermeasures.
Task Progress:	TASK DESCRIPTION The objectives of this application are to characterize the impact of space flight on female mouse retinal vasculature and tissue remodeling and to study the role of mitochondria in regulating oxidative stress-induced cellular damage and cell-cell interactions in the function of blood-retina-barrier (BRB). Our study will also determine the extent to which an antioxidant metalloporphyrin protects against space flight environment-induced changes in retinal neurovascular coupling and retinal function. FINAL REPORT The mice were launched on SpaceX-24 in December 2021 and a subset of the mice returned to Earth on January 24, 2022. The mice received weekly antioxidant or saline treatments during their stay aboard the International Space Station (ISS). Some of the mice from this return group were sent to Loma Linda University to readapt to Earth's gravity for 4 months. Tissue collection and data analysis is ongoing. Fixed ocular and brain samples will be used for histochemical analysis for identifying biomarkers for oxidative stress, apoptosis, and blood-retina-barrier (BRB) function. Frozen retina and brain samples will be used for genomics, proteomics, and metabolomics analysis. Preliminary electroretinogram (ERG) data showed that amplitude of the B-wave elicited a light-invoked slow cornea-positive potential; the amplitude was significantly decreased by 21% and 16.5% by space flight compared to habitat ground controls in the saline and BuOE treated groups respectively ($p<0.05$). The data indicate, for the first time, space flight conditions may induce bipolar cell damage and dysfunction. Our data will be presented at several national and international meetings. The data will also be presented at the upcoming American Society for Gravitational and Space Research (ASGSR) meeting.
Bibliography Type:	Description: (Last Updated: 07/10/2026)
Abstracts for Journals and Proceedings	Mao XW, Stanbouly S, Jones T, Nelson G. "Ocular response in the retina and optic nerve head after single and fractionated low-dose high-energy protons." 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. Abstracts. 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. , Feb-2022
Abstracts for Journals and Proceedings	Mao XW, Holley J, Stanbouly S, Pecaut MJ, Willey JS, Delp M. "Evaluate spaceflight induced-neuroinflammation in the mouse brain using a novel digital gene expression profiling analysis." 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. Abstracts. 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. , Nov-2021
Abstracts for Journals and Proceedings	Mao XW, Stanbouly S, Holley J, Pecaut M, Willey J, Delp M. "Biological and genomic analysis of spaceflight on neurovascular remodeling and BBB integrity." Committee on Space Research (COSPAR) 2022, 44th Scientific Assembly, Athens, Greece, July 16-24, 2022. Abstracts. Committee on Space Research (COSPAR) 2022, 44th Scientific Assembly, Athens, Greece, July 16-24, 2022. , Jul-2022

Articles in Peer-reviewed Journals	Holley JM, Stanbouly S, Pecaut MJ, Willey JS, Delp M, Mao XW. "Characterization of gene expression profiles in the mouse brain after 35 days of spaceflight mission." npj Microgravity. 2022 Aug 10;8:35. https://doi.org/10.1038/s41526-022-00217-4 ; PMID: 35948598 PMCID: PMC9365836 , Aug-2022
Articles in Peer-reviewed Journals	Mao XW, Stanbouly S, Chieu B, Sridharan V, Allen AR, Boerma M. "Low dose space radiation-induced effects on the mouse retina and blood-retinal barrier integrity." Acta Astronaut. 2022;199:412-19. https://doi.org/10.1016/j.actaastro.2022.07.029 , Jul-2022
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